

Partial differentiation.

$$y = f(x)$$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$z = f(x, y)$$

$$p = \frac{\partial z}{\partial x} = z_x = f_x = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$q = \frac{\partial z}{\partial y} = z_y = f_y = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

• 1st order partial derivatives

• 2nd order partial derivatives

$$\frac{\partial^2 z}{\partial x^2} = z_{xx} = f_{xx} = \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial x} \right] = r$$

Same numerical answers.

$$\left[\begin{array}{l} \frac{\partial^2 z}{\partial y \partial x} = z_{yx} = f_{yx} = \frac{\partial}{\partial y} \left[\frac{\partial z}{\partial x} \right] \\ \frac{\partial^2 z}{\partial x \partial y} = z_{xy} = f_{xy} = \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial y} \right] \end{array} \right] = s$$

$$\boxed{f_{xy} = f_{yx}}$$

$$\frac{\partial^2 z}{\partial y^2} = z_{yy} = f_{yy} = \frac{\partial}{\partial y} \left[\frac{\partial z}{\partial y} \right] = t$$

• Total no. of 1st order derivatives = $2^1 = 2$

• Total no. of 2nd order partial derivatives for $z = f(x, y)$ is $2^2 = 4$.

• Hence, if $z = f(x_1, x_2, \dots, x_m)$ where $x_1, x_2, \dots, x_m$ are independent variables then total no. of n^{th} order are m^n .

• If $z = f(x, y)$ then total no. of partial derivatives of n^{th} order are 2^n .

Q: If $z(x+y) = x^2 + y^2$, find $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2$

$$z = \frac{x^2 + y^2}{x+y}$$

$$\frac{\partial z}{\partial x} = \frac{2x(x+y) - 1(x^2 + y^2)}{(x+y)^2}$$

$$= \frac{x^2 - y^2 + 2xy}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{2y(y+x) - 1(y^2 + x^2)}{(y+x)^2}$$

$$= \frac{y^2 - x^2 + 2xy}{(x^2 + y)^2}$$

$$= \left[2 \frac{(x^2 - y^2)}{(x+y)^2} \right]^2$$

$$= 4 \left(\frac{x-y}{x+y} \right)^2$$

Euler's Theorem

If $z = f(x, y)$ is a homogeneous function of degree 'n'.

then $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z$

proof: as z is a homogeneous function of degree ' n '.

$$z = x^n \phi\left(\frac{y}{x}\right)$$

$$\frac{\partial z}{\partial x} = n x^{n-1} \phi\left(\frac{y}{x}\right) + x^n \phi'\left(\frac{y}{x}\right) \cdot \frac{-y}{x^2}$$

$$\frac{\partial z}{\partial y} = x^n \phi'\left(\frac{y}{x}\right) \cdot \frac{1}{x}$$

now $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$

$$= n x^n \phi\left(\frac{y}{x}\right) - x^{n-1} y \phi'\left(\frac{y}{x}\right) + x^{n-1} y \phi'\left(\frac{y}{x}\right)$$

$$= n x^n \phi\left(\frac{y}{x}\right)$$

$$= n z$$

Jacobians :-

If u, v are functions of x & y :-

$$u = u(x, y)$$

$$v = v(x, y)$$

$$\text{Then } J \begin{bmatrix} u, v \\ x, y \end{bmatrix} = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Properties :-

If u, v are functions of x, y

$$\text{Then } \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(u, v)} = 1$$

(ii) If u, v are functions of r & s and r & s are the functions of x & y :-

$$\frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(x, y)}$$

(iii) If u, v, w are not independent where u, v, w are functions of x, y, z :-
Then

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0.$$

(i) If $x = uv$; $y = \frac{u+v}{u-v}$

find $\frac{\partial(u, v)}{\partial(x, y)}$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\frac{\partial y}{\partial u} = \frac{1(u-v) - 1(u+v)}{(u-v)^2} = \frac{-2v}{(u-v)^2}$$

$$\frac{\partial y}{\partial v} = \frac{1(u-v) + 1(u+v)}{(u-v)^2} = \frac{2u}{(u-v)^2}$$

$$\Rightarrow \begin{vmatrix} u & v \\ -\frac{2v}{(u-v)^2} & \frac{2u}{(u-v)^2} \end{vmatrix}$$

$$\frac{\partial uv}{\partial (u-v)^2} = \frac{\partial (u,v)}{\partial (u,v)}$$

$$\therefore \frac{\partial (u,v)}{\partial (x,y)} = \frac{(u-v)^2}{4uv}$$

Q2 If $u = xyz$; $v = x^2 + y^2 + z^2$
 $w = x + y + z$.

find

$$\frac{\partial (x,y,z)}{\partial (u,v,w)}$$

$$\frac{\partial (u,v,w)}{\partial (x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} yz & xz & xy \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

$$\begin{aligned} & yz(2y - 2z) - xz(2x - 2y) + xy(2x - 2y) \\ & 2y^2z - 2yz^2 - 2x^2z + 2xyz + 2x^2y - 2xy^2 \\ & 2y^2(z-x) + 2x^2(y-z) - 2yz^2 + 2xz^2 \\ & \quad + 2z^2(x-y) \end{aligned}$$

$$\Rightarrow -2(y-z)(x-y)(z-x)$$

Q3 If $u = (x^2 - y^2)$; $v = 2xy$ & $x = r \cos \theta$,
 $y = r \sin \theta$ find $\frac{\partial(u, v)}{\partial(r, \theta)}$

$$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)} \quad (i)$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4x^2 + 4y^2$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r$$

from (i)

$$\frac{\partial(u,v)}{\partial(r,\theta)} = 4(x^2+y^2)r$$

$$= 4(r^2(\cos^2\theta + \sin^2\theta)r)$$

$$= 4r^3$$

$$= 4(x^2+y^2)^{3/2}$$

Q: if $u = xy + yz + zx$

$$v = x^2 + y^2 + z^2$$

$$w = x + y + z$$

whether there exists a functional relationship
b/w u, v and w and if so find it.

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} y+z & x+z & x+y \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} y+z & x+z & x+y \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$$

$$= (y+z)(y-z) +$$

$$= x^2 - z^2 - (x^2 - z^2) + x^2 - y^2$$

$$= 0$$

hence there exists a relationship b/w u, v and w

$$w^2 = v + 2u$$

Q: If $u = \frac{x+y}{1-xy}$ $v = \tan^{-1}x + \tan^{-1}y$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \frac{1(1-xy) - (-y)(x+y)}{(1-xy)^2} \quad \frac{1(1-xy) - (-2)(x-y)}{(1-xy)^2}$$

$$\frac{1-xy}{1+xy+y^2} \quad \frac{1}{1+x^2} \quad \frac{1}{1+y^2}$$

$$= \begin{vmatrix} \frac{1+y^2}{(1-xy)^2} & \frac{1+x^2}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix}$$

$$= \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2} = 0$$

$$u = \tan^{-1} u$$

Jacobians of Implicit functions :-

$$u = u(x, y)$$

$$v = v(x, y)$$

$$f_1(u, v, x, y) = 0 \quad \text{--- (i)}$$

$$f_2(u, v, x, y) = 0 \quad \text{--- (ii)}$$

partially differentiate (i) & (ii) w.r.t. x & y :-

$$\frac{\partial f_1}{\partial x} + \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial f_2}{\partial x} + \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial f_1}{\partial y} + \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial f_2}{\partial y} + \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(f_1, f_2)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_2}{\partial u} \\ \frac{\partial f_1}{\partial v} & \frac{\partial f_2}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} \frac{\partial f_1}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial f_1}{\partial v} & \frac{\partial u}{\partial x} \frac{\partial f_2}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial f_2}{\partial v} \\ \frac{\partial u}{\partial y} \frac{\partial f_1}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial f_1}{\partial v} & \frac{\partial u}{\partial y} \frac{\partial f_2}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial f_2}{\partial v} \end{vmatrix}$$

using the 4 eq's:—

$$= \begin{vmatrix} -\frac{\partial f_1}{\partial x} & -\frac{\partial f_2}{\partial x} \\ -\frac{\partial f_1}{\partial y} & -\frac{\partial f_2}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} \\ \frac{\partial f_1}{\partial y} & \frac{\partial f_2}{\partial y} \end{vmatrix}$$

$$= \frac{\partial (f_1, f_2)}{\partial (x, y)}$$

hence,

$$\frac{\partial (u, v)}{\partial (x, y)} = (-1)^2 \frac{\partial (f_1, f_2)}{\partial (x, y)} = \frac{\partial (f_1, f_2)}{\partial (u, v)}$$

Eg: If $u_1, u_2, \dots, u_n$ are functions of $x_1, x_2, \dots, x_n$

where $f_1(u_1, u_2, \dots, u_n, x_1, x_2, \dots, x_n) = 0$

$f_2(\quad \quad \quad) = 0$

$\vdots$

$f_n(\quad \quad \quad) = 0$

Then

$$\frac{\partial(u_1, u_2, \dots, u_n)}{\partial(x_1, x_2, \dots, x_n)} = (-1)^n \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}$$

$$= \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(u_1, u_2, \dots, u_n)}.$$

Q1 if $x^2 + y^2 + u^2 - v^2 = 0$
 $uv + xy = 0$

find $\frac{\partial(u, v)}{\partial(x, y)}$

$$= (-1)^2 \frac{\partial(f_1, f_2)}{\partial(x, y)}$$

$$\frac{\partial(f_1, f_2)}{\partial(u, v)}$$

$$= (-1)^2 \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} \\ \frac{\partial f_1}{\partial y} & \frac{\partial f_2}{\partial y} \end{vmatrix}$$

$$\begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_2}{\partial u} \\ \frac{\partial f_1}{\partial v} & \frac{\partial f_2}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} 2x & 2y \\ y & x \end{vmatrix} = \frac{2(x^2 - y^2)}{2(u^2 + v^2)} = \frac{x^2 - y^2}{x^2 + y^2}$$

$$\begin{vmatrix} 2u & v \\ -2v & u \end{vmatrix}$$

Q2 If $x+y+z=uvw$
 $y+z=uv$
 $z=uvw$

find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(x,y,z)}$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(u,v,w)}$$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(u,v,w)}$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x,y,z)}$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_2}{\partial u} & \frac{\partial f_3}{\partial u} \\ \frac{\partial f_1}{\partial v} & \frac{\partial f_2}{\partial v} & \frac{\partial f_3}{\partial v} \\ \frac{\partial f_1}{\partial w} & \frac{\partial f_2}{\partial w} & \frac{\partial f_3}{\partial w} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -v & -vw \\ 0 & -u & -uw \\ 0 & 0 & -uv \end{vmatrix}$$

$$= (-1)(u^2v) + u(0) - vw(0)$$

$$= -u^2v$$

$$\frac{\partial(f_1 f_2 f_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} & \frac{\partial f_3}{\partial x} \\ \frac{\partial f_1}{\partial y} & \frac{\partial f_2}{\partial y} & \frac{\partial f_3}{\partial y} \\ \frac{\partial f_1}{\partial z} & \frac{\partial f_2}{\partial z} & \frac{\partial f_3}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 1$$

$$\therefore \frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$$

Q3 If u, v, w are the roots of the eqn.

$$(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0 \quad \text{find } \frac{\partial(u, v, w)}{\partial(x, y, z)}$$

$$f_1 \Rightarrow (u - x)^3 + (u - y)^3 + (u - z)^3 = 0$$

$$f_2 (u - x)^3 + (u - y)^3 + (u - z)^3 = 0$$

$$(v - x)^3 + (v - y)^3 + (v - z)^3 = 0$$

$$(w - x)^3 + (w - y)^3 + (w - z)^3 = 0$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\partial(f_1 f_2 f_3)}{\partial(x, y, z)}$$

$$\frac{\partial(f_1 f_2 f_3)}{\partial(u, v, w)}$$

eqn $3\lambda^3 - 3\lambda^2(x+y+z) + 3\lambda(x^2+y^2+z^2) - (x^3+y^3+z^3) = 0.$

$$u+v+w = -\frac{b}{a} = +\frac{1(x+y+z)}{3}$$

$$= x+y+z$$

$$uv+vw+wu = \frac{c}{a} = \frac{2(x^2+y^2+z^2)}{3}$$

$$= x^2+y^2+z^2$$

$$uvw = -\frac{d}{a} = +\frac{1(x^3+y^3+z^3)}{3}$$

$\frac{\partial(f_1 f_2 f_3)}{\partial(x, y, z)}$	$=$	$\frac{\partial f_1}{\partial x}$	$\frac{\partial f_2}{\partial x}$	$\frac{\partial f_3}{\partial x}$
		$\frac{\partial f_1}{\partial y}$	$\frac{\partial f_2}{\partial y}$	$\frac{\partial f_3}{\partial y}$
		$\frac{\partial f_1}{\partial z}$	$\frac{\partial f_2}{\partial z}$	$\frac{\partial f_3}{\partial z}$

$$= \begin{vmatrix} -1 & -2x & -x^2 \\ -1 & -2y & -y^2 \\ -1 & -2z & -z^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2x & x^2 \\ 1 & 2y & y^2 \\ 1 & 2z & z^2 \end{vmatrix}$$

$$= 2(x-y)(y-z)(z-x)$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial w} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial w} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial w} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & v+w & vw \\ 1 & u+w & uw \\ 1 & u+v & uv \end{vmatrix}$$

$$= -2(u-v)(v-w)(w-u)$$

Partial derivatives using jacobians of implicit functions

$$u = u(x, y)$$

$$v = v(x, y)$$

$$f_1(u, v, x, y) = 0$$

$$f_2(u, v, x, y) = 0$$

partial diff^s w.r.to x :-

$$\frac{\partial f_1}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial f_1}{\partial x} \cdot 1 = 0$$

$$\frac{\partial f_2}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial f_2}{\partial x} \cdot 1 = 0$$

2 eqⁿ and 2 variables.

using cross multiplication:-

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}$$

$$\frac{\partial f_2}{\partial x} \frac{\partial f_1}{\partial v} - \frac{\partial f_2}{\partial v} \frac{\partial f_1}{\partial x} = \frac{\partial f_2}{\partial x} \frac{\partial f_1}{\partial x}$$

=

$$\frac{\partial f_1}{\partial u} \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \frac{\partial f_2}{\partial u}$$

$$\frac{\partial u}{\partial x} = \frac{\partial f_2}{\partial x} \frac{\partial f_1}{\partial v} - \frac{\partial f_2}{\partial v} \frac{\partial f_1}{\partial x} = - \frac{\partial(f_1, f_2)}{\partial(x, v)}$$

$$\frac{\partial f_1}{\partial u} \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \frac{\partial f_2}{\partial u} = \frac{\partial(f_1, f_2)}{\partial(u, v)}$$

Similarly

$$\frac{\partial v}{\partial x} = - \frac{\frac{\partial(f_1, f_2)}{\partial(u, x)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}$$

$$Q: \frac{\partial z}{\partial u} = - \frac{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, u)}}{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}}$$

Q: If $u^2 + xv^2 - xy = 0$
 $u^2 + xyv + u^2 = 0$

find $\frac{\partial u}{\partial x}$

$$\frac{\partial u}{\partial x} = - \frac{\partial(f_1, f_2)}{\partial(x, v)}$$

$$\frac{\partial(f_1, f_2)}{\partial(x, v)}$$

$$\frac{\partial(f_1, f_2)}{\partial(x, v)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} \\ \frac{\partial f_1}{\partial v} & \frac{\partial f_2}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial xv}{\partial x} \\ -2u & xy + 2v \end{vmatrix}$$

$$= (v^2 - y)xv + 2yv - 2xyv + 4uv$$

$$= v^3x - 2xyv - 2xyv + 4uv$$

$$\frac{\partial(f_1, f_2)}{\partial(x, v)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} \\ \frac{\partial f_1}{\partial v} & \frac{\partial f_2}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} v^2 - y & yv \\ 2xv & xy + 2v \end{vmatrix}$$

$$= (v^2 - y)(xy + 2v) - 2xyv^2$$

$$= v^2xy + 2v^3 - xy^2 - 2vy - 2xyv^2$$

$$=$$

Q2 If $u = x + y^2$
 $v = y + z^2$
 $w = z + x^2$

find $\frac{\partial x}{\partial u}$; $\frac{\partial y}{\partial u}$; $\frac{\partial z}{\partial u}$.

$$\frac{\partial x}{\partial u} = - \frac{\partial(f_1 f_2 f_3)}{\partial(u, y, z)} \cdot \frac{\partial(f_1 f_2 f_3)}{\partial(x, y, z)}$$

$$\frac{\partial(f_1 f_2 f_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial x} & \frac{\partial f_3}{\partial x} \\ \frac{\partial f_1}{\partial y} & \frac{\partial f_2}{\partial y} & \frac{\partial f_3}{\partial y} \\ \frac{\partial f_1}{\partial z} & \frac{\partial f_2}{\partial z} & \frac{\partial f_3}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 2x \\ 2y & 1 & 0 \\ 0 & 2z & 1 \end{vmatrix}$$

$$= 1(1) + 8xyz$$

$$= 1 + 8xyz$$

$$\frac{\partial(f_1 f_2 f_3)}{\partial(u, y, z)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_2}{\partial u} & \frac{\partial f_3}{\partial u} \\ \frac{\partial f_1}{\partial y} & \frac{\partial f_2}{\partial y} & \frac{\partial f_3}{\partial y} \\ \frac{\partial f_1}{\partial z} & \frac{\partial f_2}{\partial z} & \frac{\partial f_3}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 0 & 0 \\ 2y & 1 & 0 \\ 0 & 2z & 1 \end{vmatrix}$$

$$= -1(1) = -1$$

$$\frac{\partial x}{\partial u} = \frac{1}{1+8xyz}$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} 1 & 0 & 2x \\ -1 & 0 & 0 \\ 0 & 2z & 1 \end{vmatrix}$$

$$= -4xz$$

$$\frac{\partial y}{\partial u} = \frac{4xz}{1+8xyz}$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, u)} = \begin{vmatrix} 1 & 0 & 2x \\ 2y & 1 & 0 \\ -1 & 0 & 0 \end{vmatrix}$$

$$= 2x$$

$$\frac{\partial z}{\partial u} = \frac{-2x}{1+8xyz}$$

Taylor series :-

If $f(x, y)$ and all its partial derivatives of n^{th} order are finite and continuous for all points x and y such that $a \leq x \leq a+h$; $b \leq y \leq b+k$ then,

~~$f(a, b)$~~

$$f(a+h, b+k) = f(a, b) + \frac{1}{1!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right] f(a, b) + \frac{1}{2!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right]^2 f(a, b) + \frac{1}{3!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right]^3 f(a, b) + \dots$$

$$= f(a, b) + \frac{1}{1!} \left[h \frac{\partial f(a, b)}{\partial x} + k \frac{\partial f(a, b)}{\partial y} \right] + \frac{1}{2!} \left[h^2 \frac{\partial^2 f(a, b)}{\partial x^2} + 2hk \frac{\partial^2 f(a, b)}{\partial x \partial y} + k^2 \frac{\partial^2 f(a, b)}{\partial y^2} \right] + \frac{1}{3!} \left[h^3 \frac{\partial^3 f(a, b)}{\partial x^3} + 3h^2k \frac{\partial^3 f(a, b)}{\partial x^2 \partial y} + 3hk^2 \frac{\partial^3 f(a, b)}{\partial x \partial y^2} + k^3 \frac{\partial^3 f(a, b)}{\partial y^3} \right] + \dots$$

$$h = x - a \quad \text{and} \quad k = y - b.$$

Q1) Expand $f(x, y) = e^x \sin y$

Ans) expanding about $(0, 0)$.

$$h = (x - 0) \quad y = (y - 0).$$

$$f(x, y) = e^x \sin y + [x e^x \sin y + y e^x \cos y]$$

~~$$+ \frac{1}{2} [x^2 e^x \sin y + 2xy e^x \cos y - y^2 e^x \sin y]$$~~

~~$$+ \frac{1}{6} [x^3 e^x \sin y + 3x^2 y]$$~~

at $x=0, y=0$

$$f(x, y) = e^x \sin y \quad 0$$

$$f_x = e^x \sin y \quad 0$$

$$f_y = e^x \cos y \quad 1$$

$$f_{xx} = e^x \sin y \quad 0$$

$$f_{xy} = e^x \cos y \quad 1$$

$$f_{yy} = -e^x \sin y \quad 0$$

$$f_{xxx} = e^x \sin y \quad 0$$

$$f_{xyy} = -e^x \sin y \quad 0$$

$$f_{yyy} = -e^x \cos y \quad -1$$